Budget Internalization

Fixed-Budget RLVR Trades Test-Time Scaling for In-Budget Accuracy

Ilia Zavidnyi · Alexander Kovrigin · Arina Puchkova · Kirill Gelvan · Sergey Titov

JETBRAINS

KEY FINDING

A training cap can become a learned constraint.

Fixed-budget training improves near-cap accuracy, but can remove the ability to scale with more test-time tokens.

+38.9 pp at 4k tokens

−35.0 pp at 64k tokens

Qwen 4k-trained vs. base; AIME avg@3

00 The experiment

Compare base and trained models below and above the training cap.

TRAINING PROTOCOL

GRPO on DeepScaleR math, replicated on Reasoning Gym. Reward 1 for correct completed answers; 0 for incorrect or truncated rollouts.

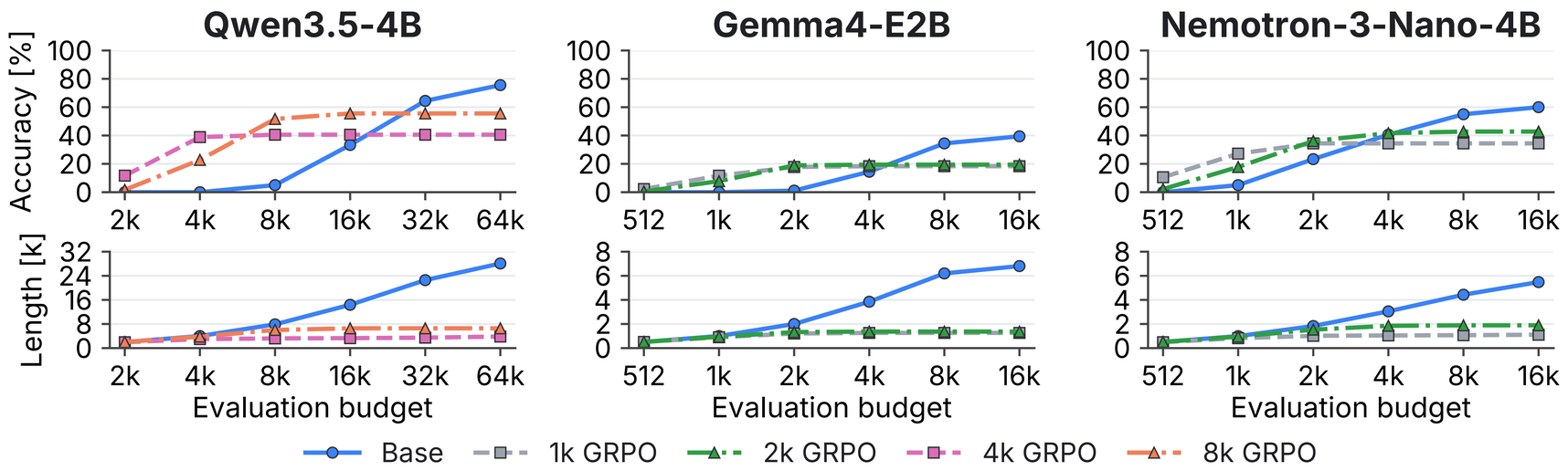
MATH TRAINING CAPS

Qwen3.5-4B: 4k / 8k.
Gemma4-E2B: 1k / 2k.
Nemotron-3-Nano-4B: 1k / 2k.

EVALUATION

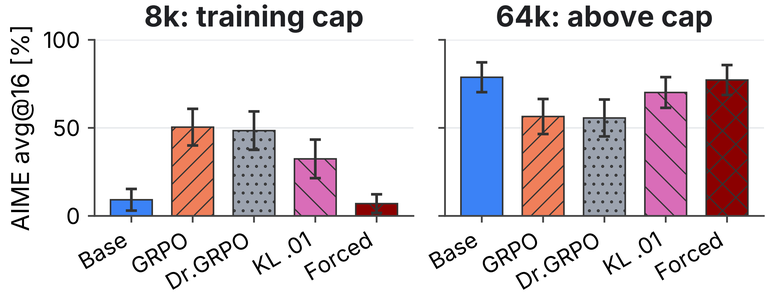
AIME 2024/2025 (60), GPQA-Diamond (198), LiveCodeBench v6 (175).

01 Better near the cap; weaker beyond it



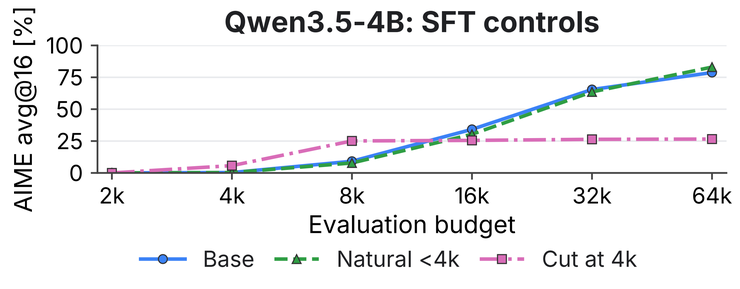
Extra allowance does not produce longer responses in trained models. Across six math checkpoints: +11.7 to +46.7 pp at the cap, then −17.2 to −35.0 pp at the largest shown budget. Length and accuracy saturate beyond the cap. AIME avg@3; lengths are means. Legend labels denote GRPO caps.

02 What preserves scaling?



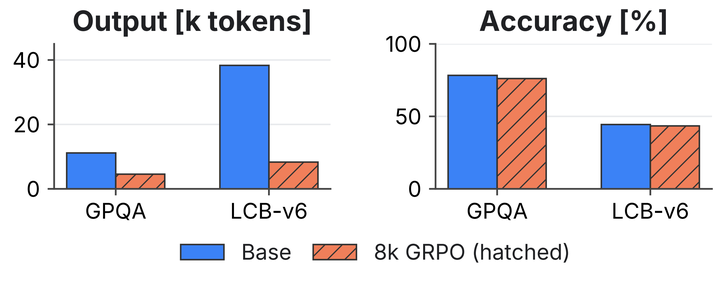
Qwen, 8k math training; avg@16 on 60 problems. Error bars: 95% CIs. Dr.GRPO does not restore scaling. Strong KL and forced answers recover high-budget accuracy at the expense of near-cap gains. Hatching marks post-training.

03 The effect extends to SFT



SFT uses successful self-generated traces: natural traces finish within 4k; cut traces stop at 4k and receive the correct answer. At 64k, accuracy is 83.0% vs. 26.6%. Artificial shortening also limits SFT scaling.

04 Shorter can be useful



At 64k (avg@3), Qwen's 8k math-trained checkpoint uses 59% fewer GPQA tokens and 78% fewer LCB tokens, losing only 2.2 and 1.0 pp accuracy. Compression helps when extra reasoning brings little benefit.

TAKEAWAY **Evaluate accuracy and response length at multiple budgets.**

Scope: three small model families; one training seed per condition. avg@k averages k samples per problem; controls use avg@16.

WEBSITE
zavidnyi.com//
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